

MSc Data Science Project

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Department of Physics, Astronomy and Mathematics

Data Science FINAL PROJECT REPORT

Project Title:

**Predicting Customer Subscription Behaviour Using Supervised
Classification Techniques**

By

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GitHub Link:

[https://github.com/JISOOUNNIE1234/MSc Data Science Project
.git](https://github.com/JISOOUNNIE1234/MSc_Data_Science_Project.git)

Supervisor: Dr Ashley Spindler

Word Count: 4980

Declaration Statement

I am submitting this report to partially fulfill the requirements for the Master of Science in Data Science degree at the University of Hertfordshire.

I confirm that I have carefully reviewed the university's guidelines regarding academic integrity, plagiarism, and assessment offenses. I fully understand how the university handles suspected academic misconduct and am aware that potential penalties include failing this project or the entire course.

I declare that this submission is entirely my own original work. Any ideas, data, or quotes sourced from the published or unpublished work of others have been properly cited and fully acknowledged in accordance with university regulations (Ref. UPR AS/C/6.1, section 7 and UPR AS/C/5, section 3.6).

Furthermore, I confirm that no human participants were involved in the research for this MSc project.

Finally, I grant permission for this report to be shared on university module websites, provided that my authorship is clearly acknowledged.

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Acknowledgement

This project is a major milestone in my education and demonstrates the practical and analytical skills I have built during my coursework. Completing this research allowed me to step outside of standard classroom theory, apply my knowledge to real-life situations, and develop a much stronger and clearer understanding of my field.

I want to share my deep gratitude with my supervisor, Ashley Spindler, for his steady help, wise advice, and helpful critiques from start to finish. His encouragement, combined with the excellent tools and learning space offered by the University of Hertfordshire, were absolutely essential in guiding the direction and overall quality of this research.

Finally, I also want to recognize my own hard work, self-control, and ongoing effort in finishing this study. Keeping my focus, pushing past obstacles, and remaining dedicated to learning every step of the way were the main reasons I was able to successfully complete this project.

Abstract

This study focuses on using supervised machine learning to forecast if a bank customer will sign up for a new service. The research is built on the Bank Marketing dataset, which contains over 45,000 entries and 17 different details about each person, such as their background, financial health, and past marketing interactions. The main goal is to predict whether a customer will eventually agree to a term deposit.

The project followed a clear, step-by-step path that included cleaning the data, looking for early trends, adjusting the variables, and creating the models. A major challenge was that the data was heavily lopsided, as only about 11.7% of the people actually subscribed. Because of this uneven split, basic accuracy was not a trustworthy way to measure success. Instead, the study used specialized mathematical tests-such as precision, recall, F1-score, and ROC-AUC-to fairly judge the models. To get the data ready, text-based categories were turned into number codes, and all numerical values were leveled out so the computer could read them equally.

Several different predictive models were built and compared, specifically Logistic Regression, Decision Tree, and Random Forest. Out of these choices, the Random Forest model performed the best, offering the most reliable balance across all the mathematical tests. To further fix the lopsided data issue and make the model smarter, specific techniques were used to balance the importance of the different customer groups, such as adjusting their weights and reducing the size of the larger group (undersampling).

Overall, the findings prove that carefully preparing the data and picking the correct algorithms are essential for making trustworthy predictions. By using this final model, banks can create much smarter marketing plans. It allows them to focus their energy only on the customers who are most likely to say yes, which boosts the success rate of their campaigns and prevents them from wasting money on unnecessary advertising.

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1 Introduction:

In the banking industry, companies frequently run marketing campaigns-often through direct phone calls-to sell financial products like term deposits. However, calling everyone is both expensive and time-consuming, especially since most people simply say no. Because of this, banks need smarter, data-driven ways to guess which customers will actually buy the product. By identifying the people who are most likely to say yes, banks can focus their efforts, save money, and make their campaigns much more successful. Customer segmentation helps banks group individuals based on behaviour and preferences, enabling more targeted marketing strategies, as illustrated in Figure 1.

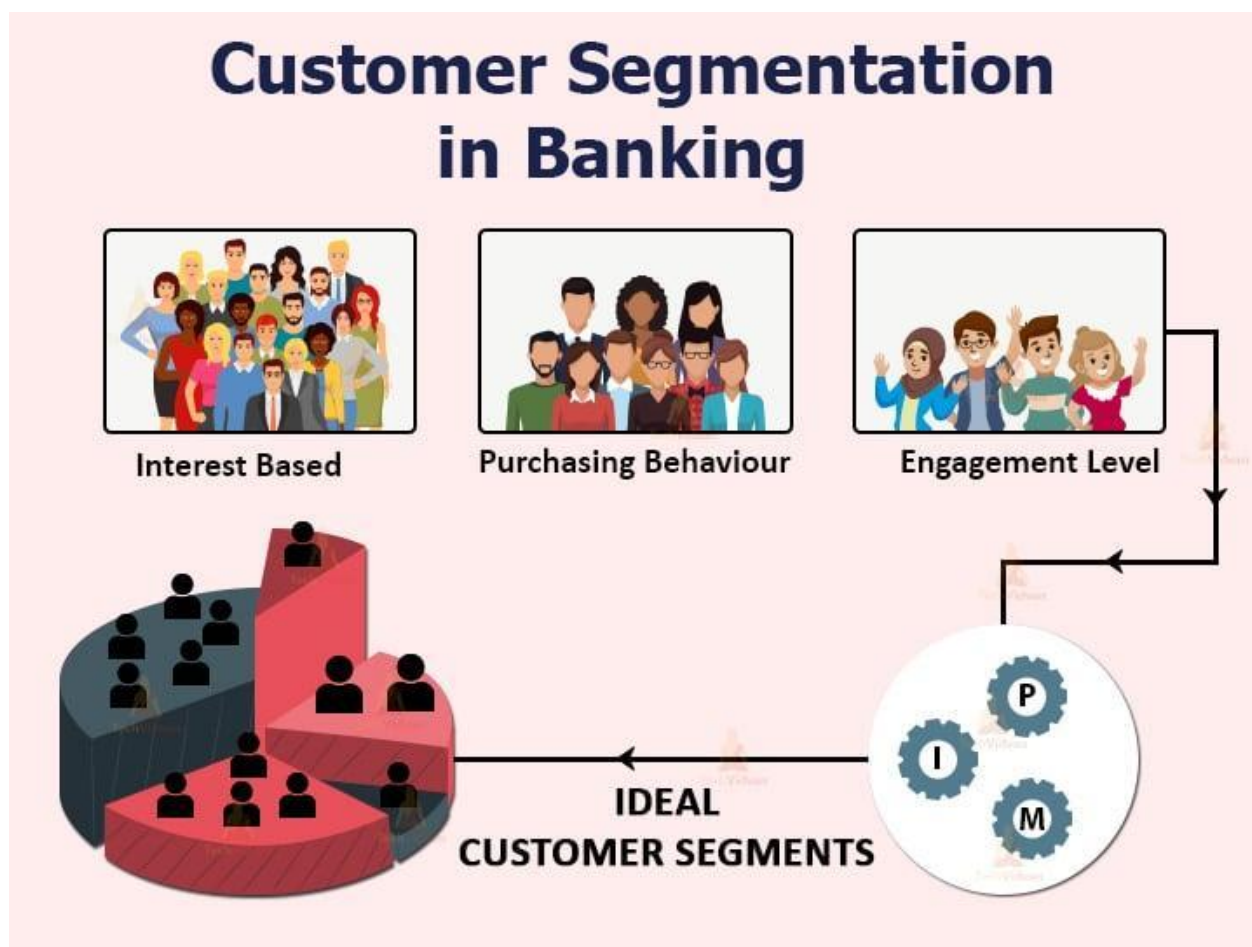


Figure 1: Customer segmentation in banking based on customer characteristics (source from Muguro, n.d.)

Predicting whether a customer will sign up is known as a supervised classification task. This means the computer uses past records to learn how to predict a simple "yes" or "no" answer for new customers. This project uses the Bank Marketing dataset, which

tracks direct phone campaigns run by a Portuguese bank between 2008 and 2013. The data contains 45,211 records with 17 different details about each person, including their age, job, education, bank balance, and how long they spoke on the phone. The main goal is to predict the final outcome: did the customer subscribe to the deposit?

Past research proves that machine learning is an excellent tool for understanding customer habits. For example, Moro, Cortez, and Rita (2014) built a system to predict the success of bank phone campaigns, proving that these computer models can uncover highly valuable trends in customer data. Standard models, like Logistic Regression and Decision Trees, are often used as a starting point because they are simple and easy to understand. However, more advanced combined tools, like Random Forest, generally do a much better job of handling complicated data (Breiman, 2001).

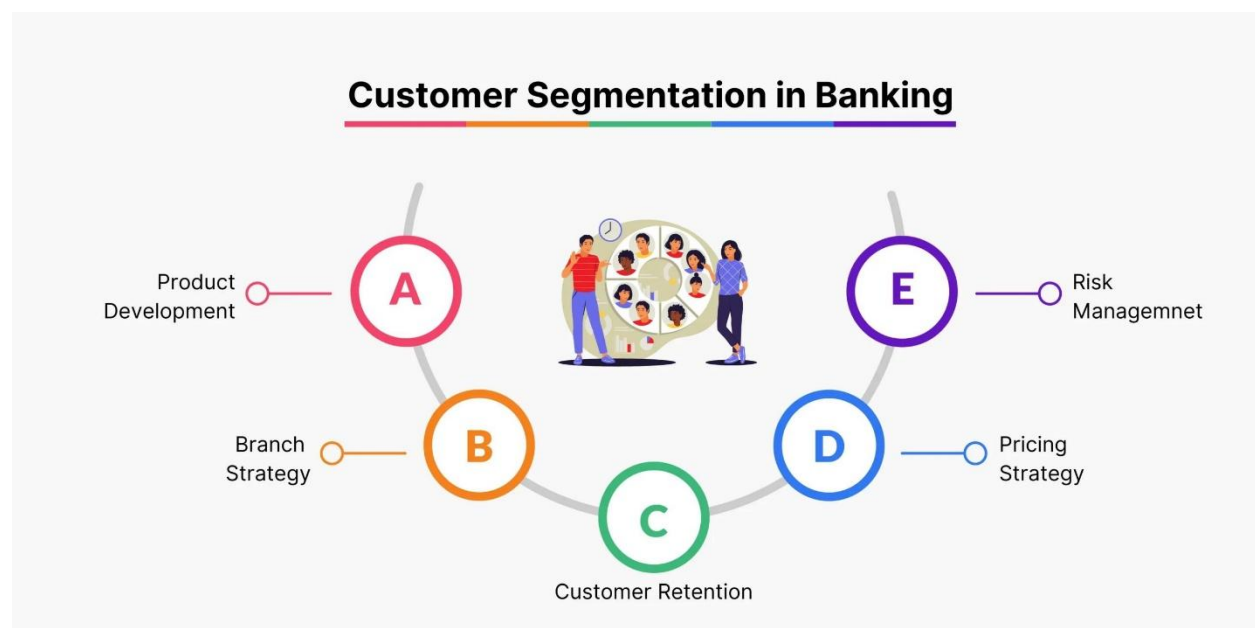


Figure 2: Applications of customer segmentation in banking, (source from QServices IT, n.d.).

A major challenge with this specific dataset is that it is heavily lopsided (class imbalance). Very few people actually subscribed compared to the massive number of people who declined. If a model just guesses "no" every time, it might still score a high "accuracy" rating, which is misleading. To properly judge how well the models are working on this uneven data, the study uses specialized mathematical tests instead, such as precision, recall, F1-score, and ROC-AUC.

Cleaning the data before testing is also a mandatory step. Real-world data is usually messy, containing blank spots or text that computers cannot process. In this project, missing details labeled as "unknown" were carefully handled, text-based categories were turned into number codes (a process called one-hot encoding), and all numerical

values were leveled out to be on the same scale. These cleaning steps guarantee that the computer can learn from the data without getting confused

The primary goal of this project is to build and compare several different predictive models including Logistic Regression, Decision Tree, and Random Forest to find the most reliable way to forecast customer sign-ups. To fix the issue of lopsided data, the study uses techniques like adjusting the importance of the smaller group (class weighting) and shrinking the size of the larger group (undersampling). The project follows a strict step-by-step path, from exploring the initial data to grading the final models.

Ultimately, this research demonstrates how modern machine learning can be applied to real banking data to make smarter business choices. The results can help financial companies improve their marketing by focusing only on the people who are likely to subscribe, cutting out wasted effort and increasing overall profits.

2 Literature Review:

Anticipating whether a client will sign up for a service is a central goal of direct marketing. Financial institutions specifically need to identify individuals most likely to open a term deposit. Research frequently relies on the Bank Marketing dataset, which tracks the promotional efforts of a bank in Portugal. This collection includes both personal details and campaign metrics, serving as the foundation for predicting final enrollment.

Past investigations confirm that machine learning is highly effective for these forecasts. For instance, Moro, Cortez, and Rita (2014) developed a system to evaluate telemarketing success, demonstrating that supervised learning can uncover valuable client trends. Their findings highlight how historical records can refine promotional tactics and guide smarter business choices. Class imbalance occurs when one class significantly outweighs the other, which can negatively affect model performance, as shown in Figure 3

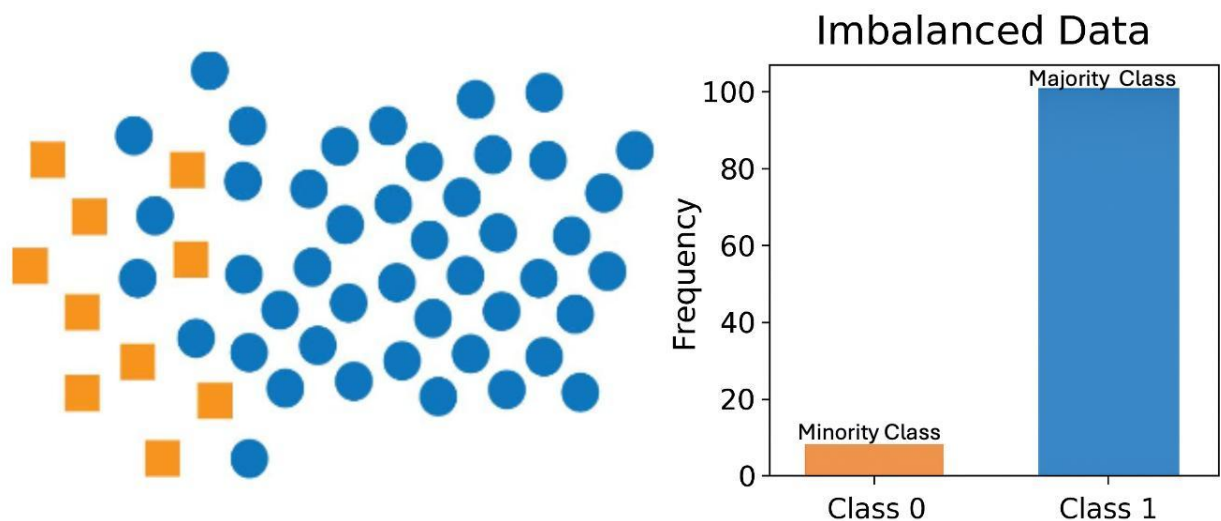


Figure 3: Illustration of class imbalance showing the dominance of the majority class over the minority class in a dataset (source from AI Learning Hub, n.d.).

A foundational step in any machine learning project is exploring the data to understand its shape and spot any unusual features. In marketing records, a frequent hurdle is uneven data distribution, meaning the number of people who decline a product drastically outnumbers those who accept it. This unevenness can ruin model training because basic accuracy scores will hide the model's true flaws. Consequently, researchers must rely on specialized evaluation tools and data adjustments to handle this imbalance effectively.

Cleaning the information is also crucial for building strong models. Real-world data is frequently messy or incomplete. In this specific dataset, empty fields are marked as

"unknown," which must be managed carefully to prevent skewed results. A standard fix is to treat these spots as missing and fill them with the most common answer, keeping the dataset intact while offering a simple solution.

Another necessary preparation step is changing text-based labels into number codes. Because machine learning tools run on mathematics, methods like one-hot encoding are used to flip these text categories into simple binary numbers. While this creates more data columns, it allows the algorithms to accurately read the categories. Number-based data is also leveled out to guarantee that every detail carries the same weight during the learning phase. Imbalanced datasets can lead to biased decision boundaries, where the model favours the majority class, as illustrated in Figure 4

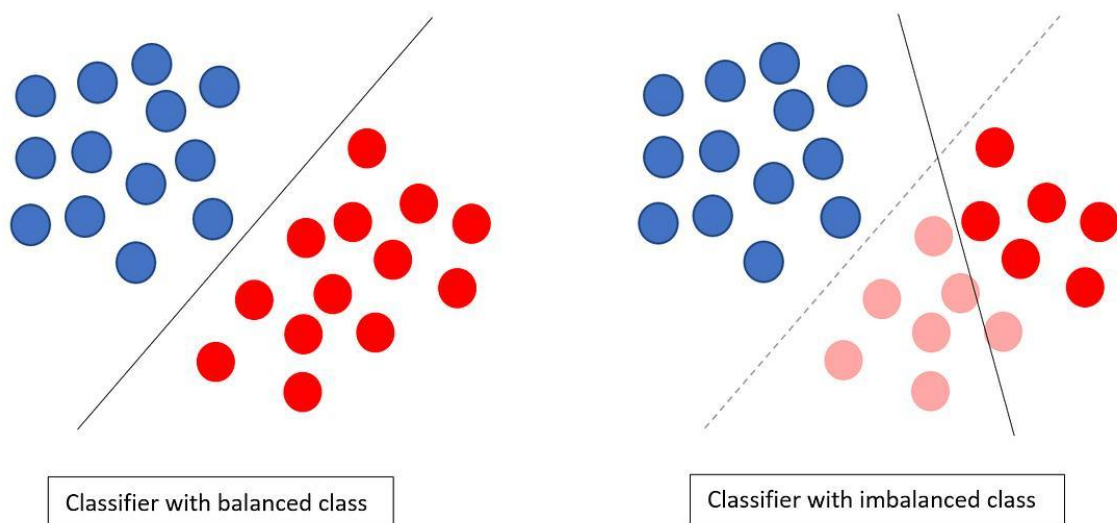


Figure 4: Comparison of classification boundaries in balanced and imbalanced datasets, (adapted from X8 AI Community, n.d.).

To ensure the final tests are fair, the information is divided into training and testing groups. When dealing with uneven data, researchers use a method called stratified splitting. This guarantees that the original ratio of "yes" and "no" answers stays exactly the same in both groups, leading to much more trustworthy test results

Several predictive tools can be used for this challenge. Logistic Regression is often chosen as a starting point because it is straightforward to understand. Decision Trees provide a better way to map out complicated, twisting data paths, while Random Forest improves overall accuracy by merging many individual trees together. Random Forest is especially strong with standard data tables because it can map out overlapping

connections (Breiman, 2001). Furthermore, techniques that assign more importance to the smaller group of "yes" answers can be used to balance the learning process.

Judging models on uneven data requires specific grading tools. Relying purely on basic accuracy is dangerous, as a model could score highly just by blindly guessing the majority answer every time. Therefore, researchers use advanced scores like precision, recall, F1-score, and ROC-AUC for a proper review. Analyzing precision and recall is especially valuable when successful sign-ups are rare, as it measures the model's actual ability to find those specific clients.

Another major risk in this research area is data leakage. Certain details, such as how long a phone call lasted, are excellent predictors but are impossible to know before the call actually happens. Using these details to train the model can create falsely perfect results. Therefore, researchers must be very careful to only include information that would actually be available in a real-world scenario.

In summary, prior studies confirm that successfully forecasting client behavior requires thorough data cleaning, smart solutions for uneven group sizes, and the right machine learning tools. Combined systems like Random Forest consistently show the best results, provided that researchers use the correct mathematical tests to prove their true value.

3 Methodology:

This section explains the steps taken to build computer models that guess if a bank customer will sign up for a service. The project followed a clear path: planning the system, learning about the data, cleaning it, finding early trends, adjusting the details, and building the models. This careful, step-by-step process makes sure the final results are accurate and trustworthy.

3.1 System Overview

The project uses a standard machine learning setup. First, the data is collected and studied. Next, it is cleaned up so the computer algorithms can read it properly. After that, an initial review known as Exploratory Data Analysis is done to see how the information is spread out and how different details connect to one another.

Once the data is clear, certain details are adjusted to make them more useful. The information is then split into two groups: one for training the computer and one for testing it. Finally, several different models are built and compared to see which one makes the best predictions.

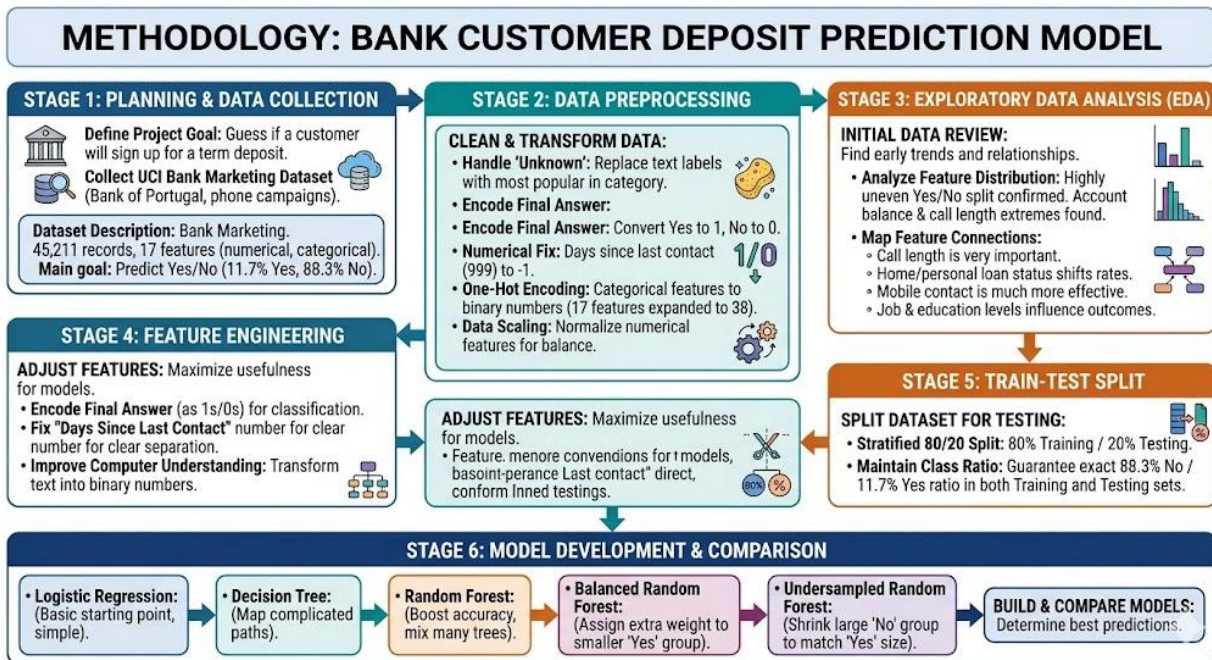


Figure 5: Methodology - System Overview

3.2 Dataset Description

This study uses the Bank Marketing dataset from the UCI Machine Learning Repository. It holds 45,211 records and 17 different details about phone marketing campaigns run by a bank in Portugal.

The data includes both numbers (such as age, account balance, call length, and number of past contacts) and text categories (such as job type, marital status, education level, and loan status). The main goal is to predict a simple yes or no: did the customer agree to the term deposit?

While there are no completely blank spots, some text categories contain the word "unknown." Also, the data is heavily unbalanced. About 88.3% of the people in the records said no to the offer, while only 11.7% said yes.

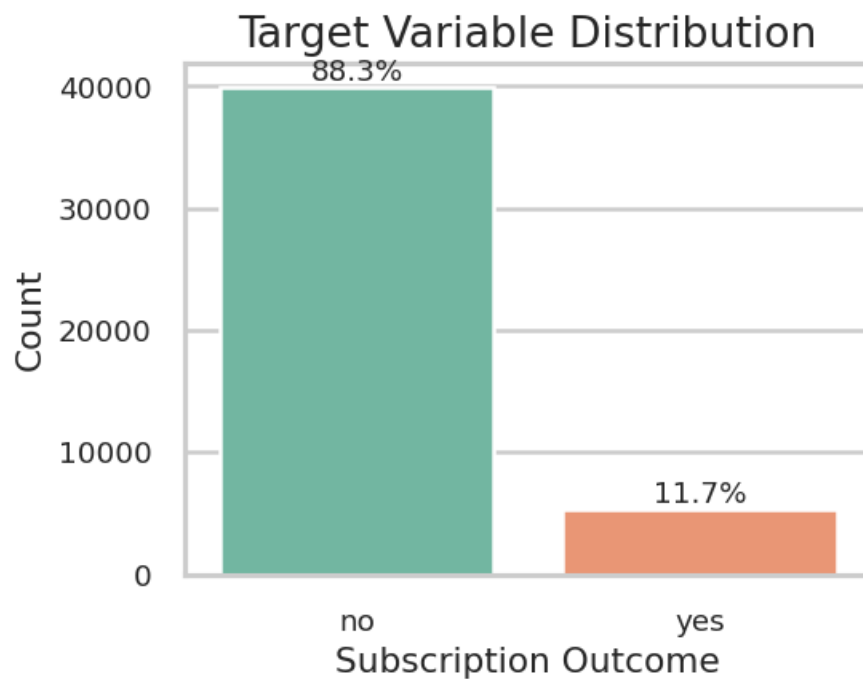


Figure 6: Target Variable Distribution

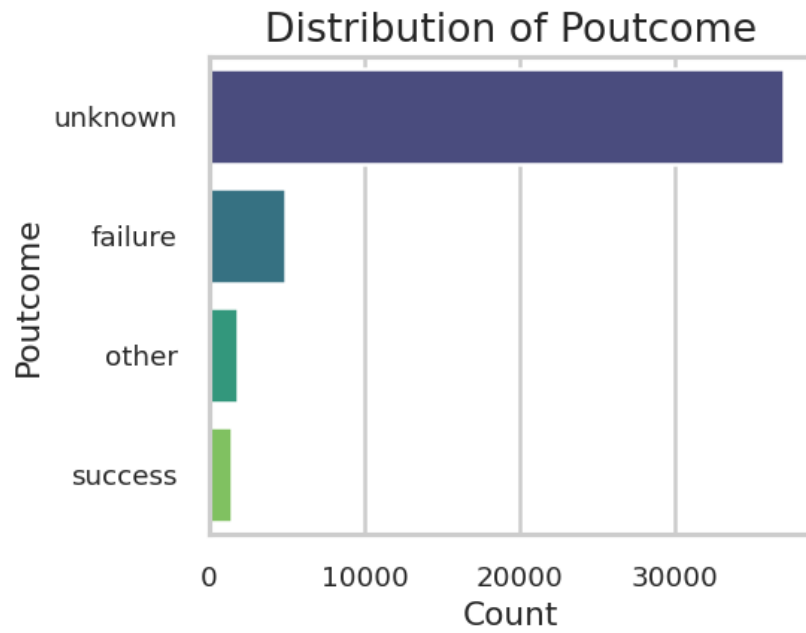


Figure 7: Distribution of Performance

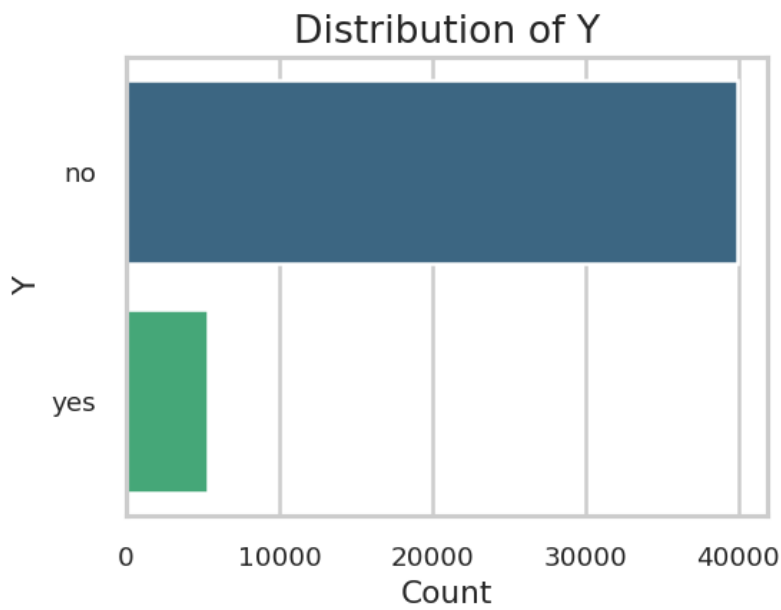


Figure 8: Distribution of Y

3.3 Data Preprocessing

The data had to be cleaned to make it ready for the models.

First, the "unknown" labels were treated as missing information and replaced with the most popular answer in that specific category.

The final "yes or no" answer was changed into numbers: 1 for yes and 0 for no.

One specific detail, which tracked the days since the last contact, used the number 999 to show a person had never been called before. This was changed to -1 to make more sense mathematically.

Text categories were turned into numbers using a method called one-hot encoding. This expanded the total number of data features from 17 to 38.

Finally, all the number-based details were scaled so they were on the same level, preventing larger numbers from unfairly taking over the model.

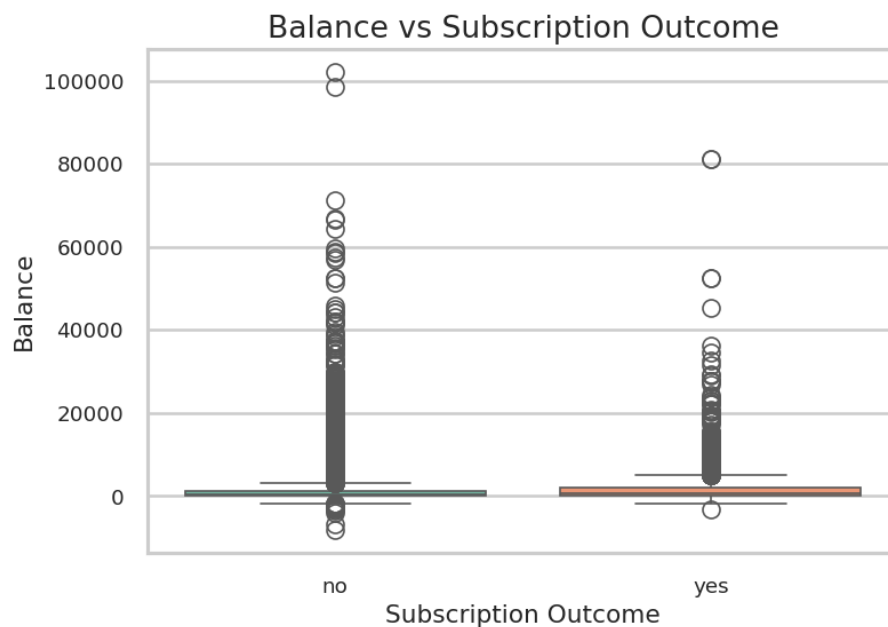


Figure 9: Balance vs Subscription Outcome

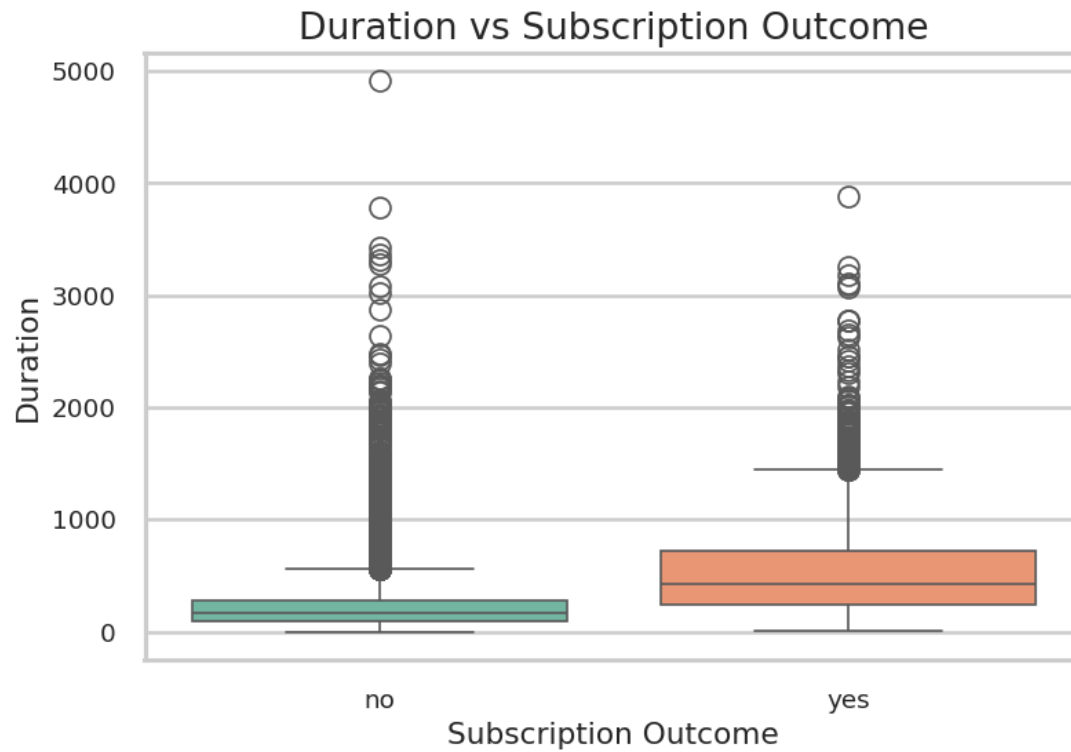


Figure 10: Duration vs Subscription Outcome

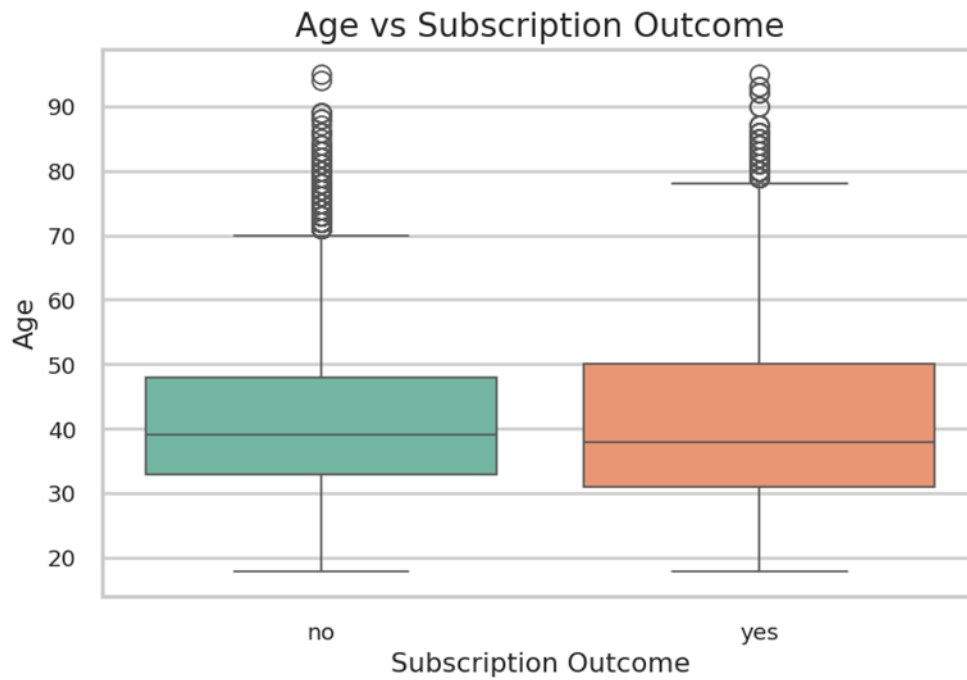


Figure 11: Age vs Subscription Outcome

3.4 Exploratory Data Analysis (EDA)

A basic review was done to find early trends in the information. The most obvious finding was the highly uneven split between people who said yes and no, proving that standard grading methods would not work later on.

Looking at the numbers showed that details like account balances and call lengths had some extreme highs. It also showed that most people had never been contacted by the bank prior to this campaign.

Looking at how things connected, the length of the phone call was very important: longer calls usually led to a "yes." Different jobs and education levels also shifted the sign-up rates. People without home or personal loans were slightly more likely to say yes. Also, calling mobile phones worked much better than other contact methods.

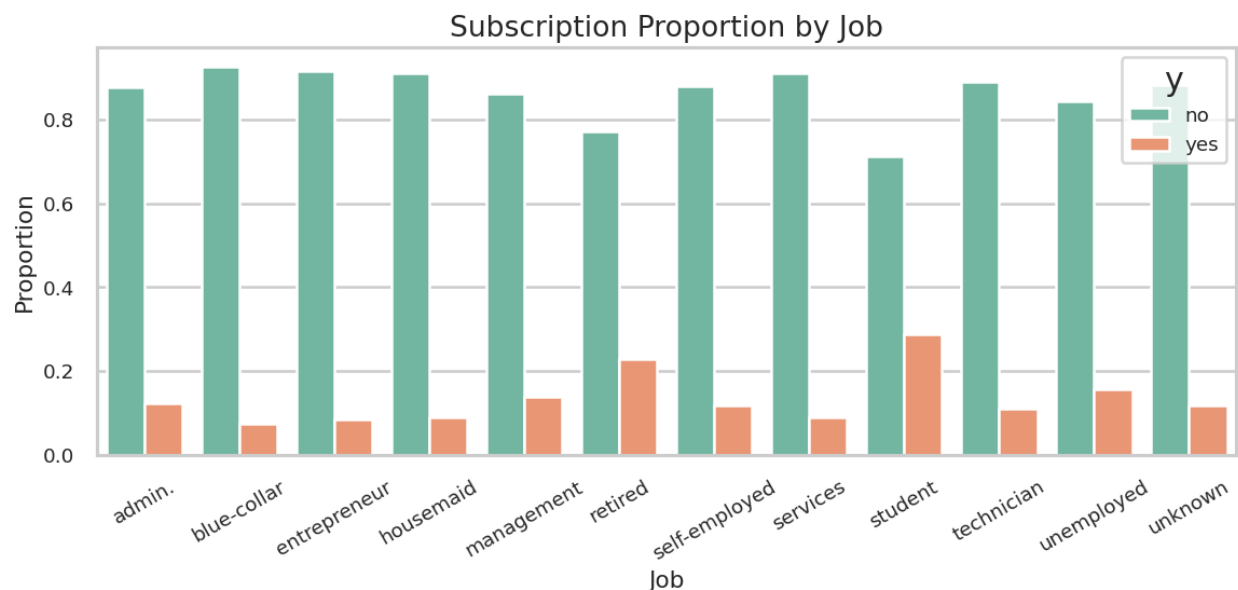


Figure 12: Subscription Proportion by Job

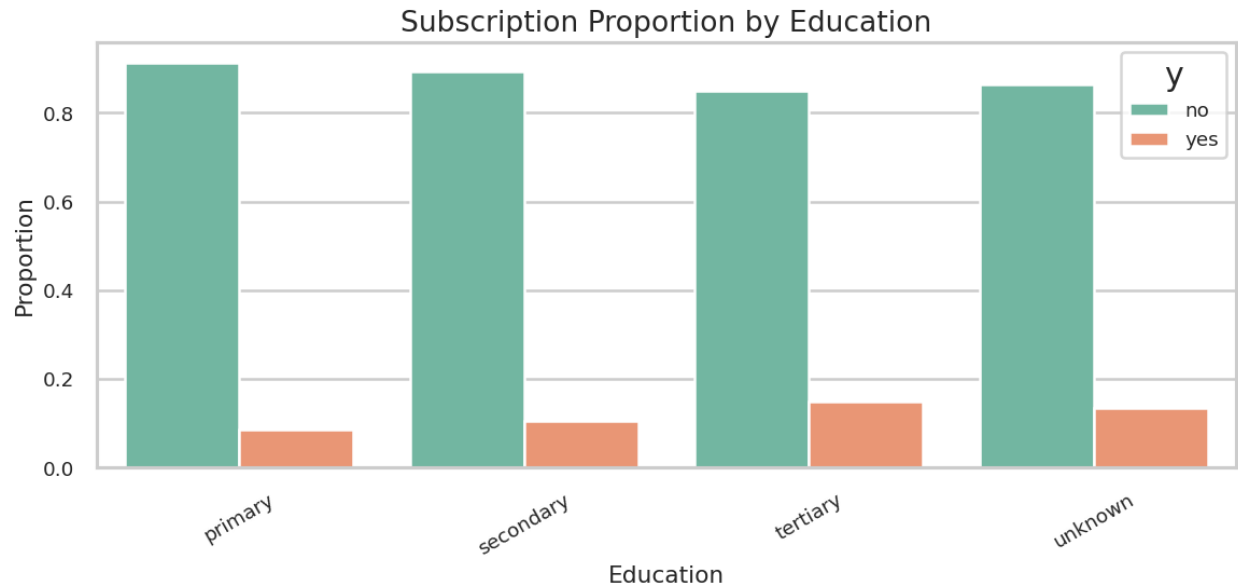


Figure 13: Subscription Proportion by Education

3.5 Feature Engineering

The data was adjusted to help the computer learn better. Changing the final answer to 1s and 0s allowed the classification models to function. Fixing the number for "days since last contact" helped clearly separate new customers from old ones. While no entirely new categories were invented, changing the text into simple binary numbers greatly improved how the computer understood the hidden relationships in the data.

3.6 Train-Test Split

To fairly test the models, the dataset was cut into two pieces: 80% for training and 20% for testing. Because the "yes" and "no" answers were so unbalanced, a special splitting method was used. This guaranteed that the 80/20 split kept the exact same ratio of 88.3% "no" and 11.7% "yes" in both the practice group and the test group.

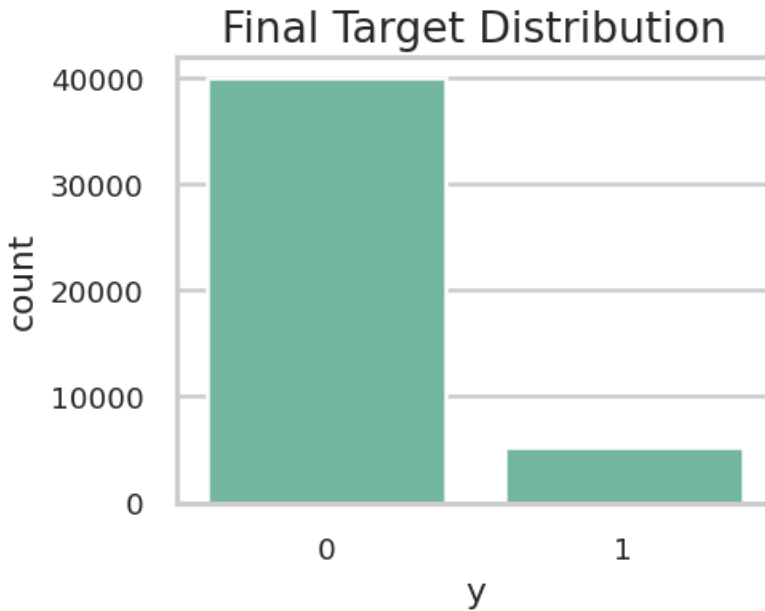


Figure 14: Final Target Distribution

3.7 Model Development

Several different models were built to compare their success.

- **Logistic Regression** was used first as a basic starting point because it is simple and easy to understand.
- A **Decision Tree** was used to map out more complicated, twisting paths in the data.
- **Random Forest** was used to boost accuracy by mixing many decision trees together.

To handle the uneven "yes" and "no" answers, a special version of Random Forest was trained to give extra weight and importance to the much smaller group of "yes" answers. Additionally, another test was run by shrinking the large group of "no" answers until it exactly matched the size of the "yes" group (a method called Random Undersampling). A Random Forest model was then trained on this perfectly balanced data to see if it improved the computer's ability to guess the rare "yes" outcomes.

4 Results

This section evaluates how well the machine learning models predicted customer sign-ups. The models were judged on their accuracy, precision, recall, F1-score, and ROC-AUC, with a specific focus on how the uneven data affected the final results.

4.1 Baseline Models Performance

The first model tested was Logistic Regression. It achieved a strong accuracy of 0.8990 and an ROC-AUC of 0.9006. However, it struggled to identify most of the actual sign-ups, resulting in a low recall of 0.3270 and an F1-score of 0.4312.

Next, the Decision Tree model was evaluated. Its accuracy fell to 0.8660 and its ROC-AUC dropped to 0.6929. While its recall improved slightly to 0.4669, its precision was poor, leading to an F1-score of 0.4491.

Finally, the Random Forest model performed the best among the starting options. It reached an accuracy of 0.9055 and an ROC-AUC of 0.9195. Most importantly, it offered the most balanced results, scoring 0.6624 in precision, 0.3913 in recall, and 0.4920 for its F1-score.

Table 1: Baseline Model Performance

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.8990	0.6325	0.3270	0.4312	0.9006
Decision Tree	0.8660	0.4326	0.4669	0.4491	0.6929
Random Forest	0.9055	0.6624	0.3913	0.4920	0.9195

4.2 Handling Class Imbalance (Class Weighting)

To fix the uneven data issue, a balanced version of the Random Forest was tested by adjusting class weights. The overall accuracy stayed high at 0.9038, and the ROC-AUC slightly increased to 0.9206. However, its ability to find the rare positive cases actually worsened. The recall dropped to 0.3318 and the F1-score fell to 0.4466. This proved that simply adjusting the class weights did not solve the problem of finding the smaller group.

Table 2: Random Forest vs Balanced Random Forest

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Random Forest	0.9055	0.6624	0.3913	0.4920	0.9195
Random Forest (Balanced)	0.9038	0.6829	0.3318	0.4466	0.9206

4.3 Handling Class Imbalance (Undersampling)

A different approach was taken by shrinking the larger group. The majority class was reduced from 31,937 records down to 4,231, making both groups perfectly equal. When a Random Forest model was trained on this new balanced data, its recall skyrocketed to 0.8639, meaning it became excellent at finding actual sign-ups. The downside was that precision dropped to 0.4004 and total accuracy fell to 0.8327. However, the F1-score reached 0.5471, the highest of all models, showing a much better trade-off between precision and recall

Table 3: Class Distribution before and after Sampling

Stage	Class 0	Class 1
Before Undersampling	31,937	4,231
After Undersampling	4,231	4,231

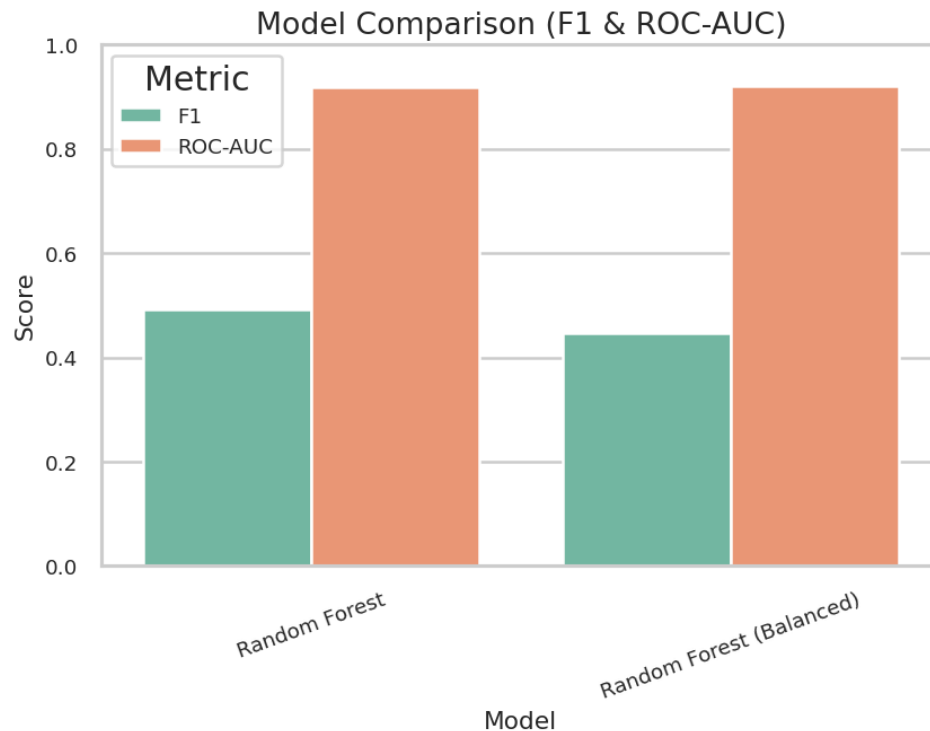


Figure 15: Model Comparison (F1 & ROC-AUC)

4.4 Final Model Comparison

Comparing all the tools shows that the standard Random Forest offers the best all-around stability. However, the undersampled Random Forest is the clear winner for achieving the highest recall and F1-score. This makes the undersampled model the best choice for businesses that care more about finding every possible positive lead rather than keeping overall accuracy perfect.

Table 4: Final Model Comparison

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.8990	0.6325	0.3270	0.4312	0.9006
Decision Tree	0.8660	0.4326	0.4669	0.4491	0.6929
Random Forest	0.9055	0.6624	0.3913	0.4920	0.9195
Random Forest (Balanced)	0.9038	0.6829	0.3318	0.4466	0.9206
Random Forest (Undersampled)	0.8327	0.4004	0.8639	0.5471	0.9174

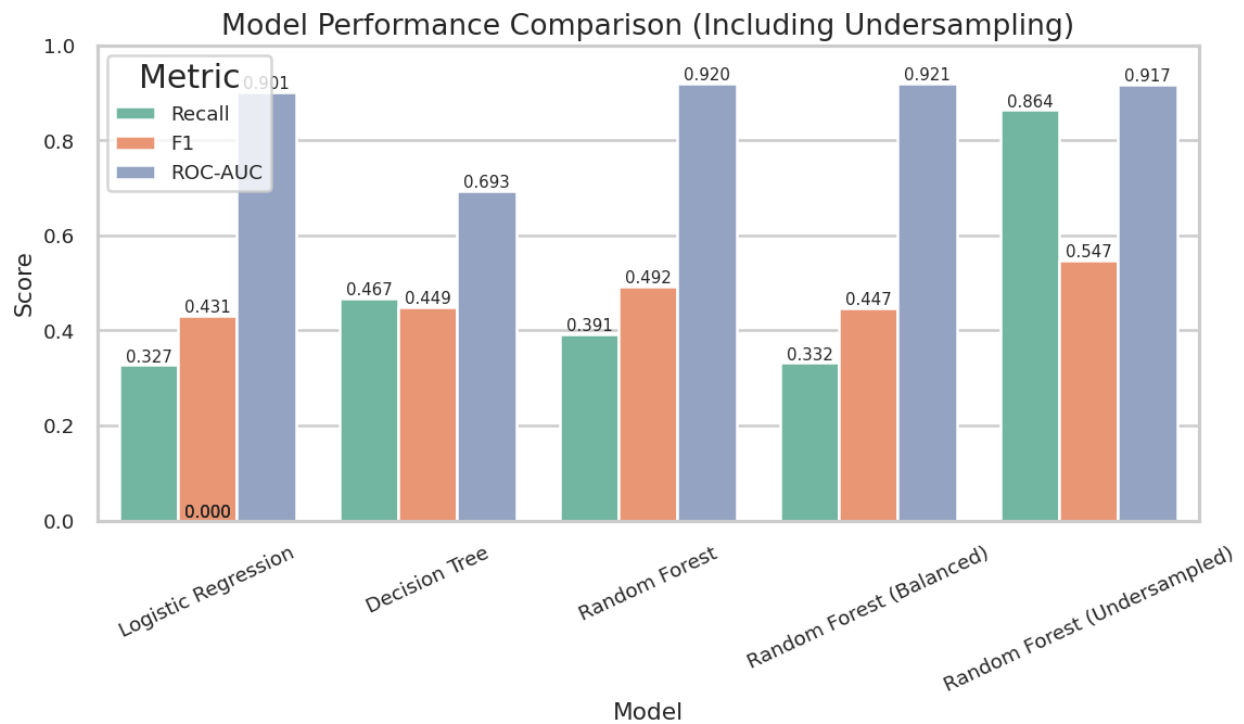


Figure 16: Model Performance Comparison (Including Understanding)

Confusion Matrix - Logistic Regression

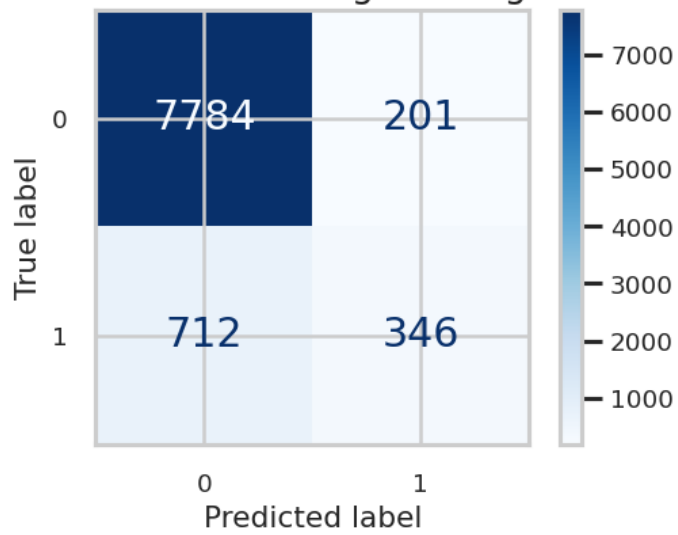


Figure 17: Confusion Matrix - Logistic Regression

Confusion Matrix - Decision Tree

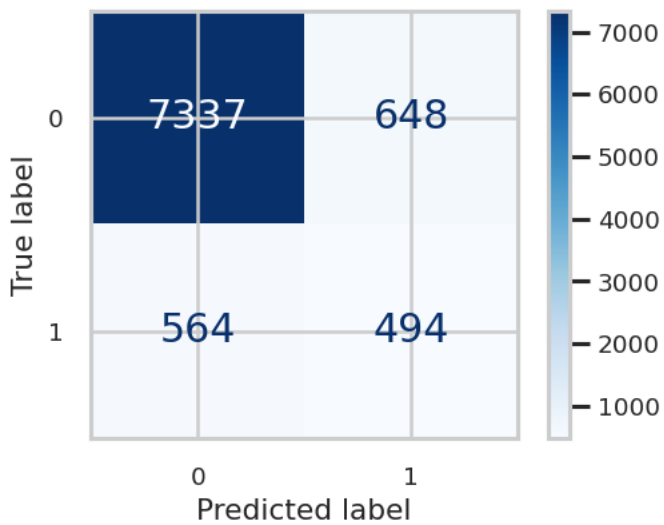


Figure 18: Confusion Matrix - Decision Tree

Confusion Matrix - Random Forest

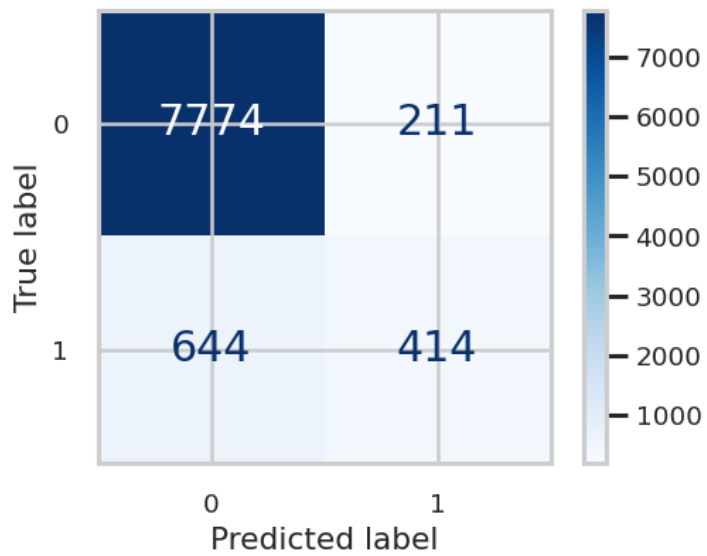


Figure 19: Confusion Matrix - Random Forest

Confusion Matrix - Random Forest (Balanced)

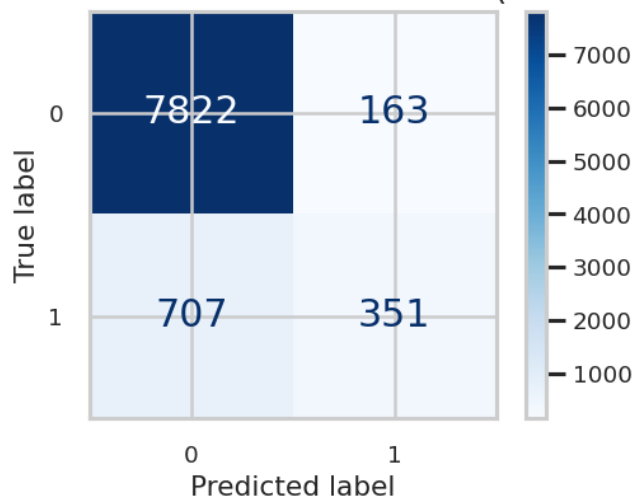


Figure 20: Confusion Matrix - Random Forest (Balanced)

Confusion Matrix - Random Forest (Undersampled)

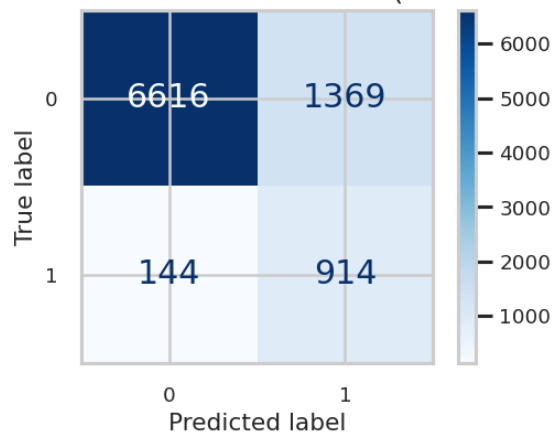


Figure 21: Confusion Matrix - Random Forest (Undersampled)

4.5 Cross-Validation Results

A standard 5-fold cross-validation test was run to make sure the model was stable on different pieces of data. The F1-scores across all five tests remained very steady, finishing with a reliable average score of 0.4105.

Table 5: Cross-Validation Performance

Fold	F1 Score
------	----------

1	0.4174
---	--------

2	0.3743
---	--------

3	0.4217
---	--------

4	0.4066
---	--------

5	0.4323
---	--------

Mean	0.4105
------	--------

4.6 Feature Importance

An analysis of the data features showed that the length of the phone call (duration) was the most powerful clue for predicting a sign-up. Other major factors included the customer's account balance, age, and the day of the contact. Details from past campaigns, such as previous success and days since the last contact, also heavily influenced the final predictions.

Table 6: Top Important Features

Rank	Feature
1	duration
2	balance
3	age
4	day
5	poutcome_success
6	pdays
7	housing_yes
8	campaign
9	previous
10	month_may

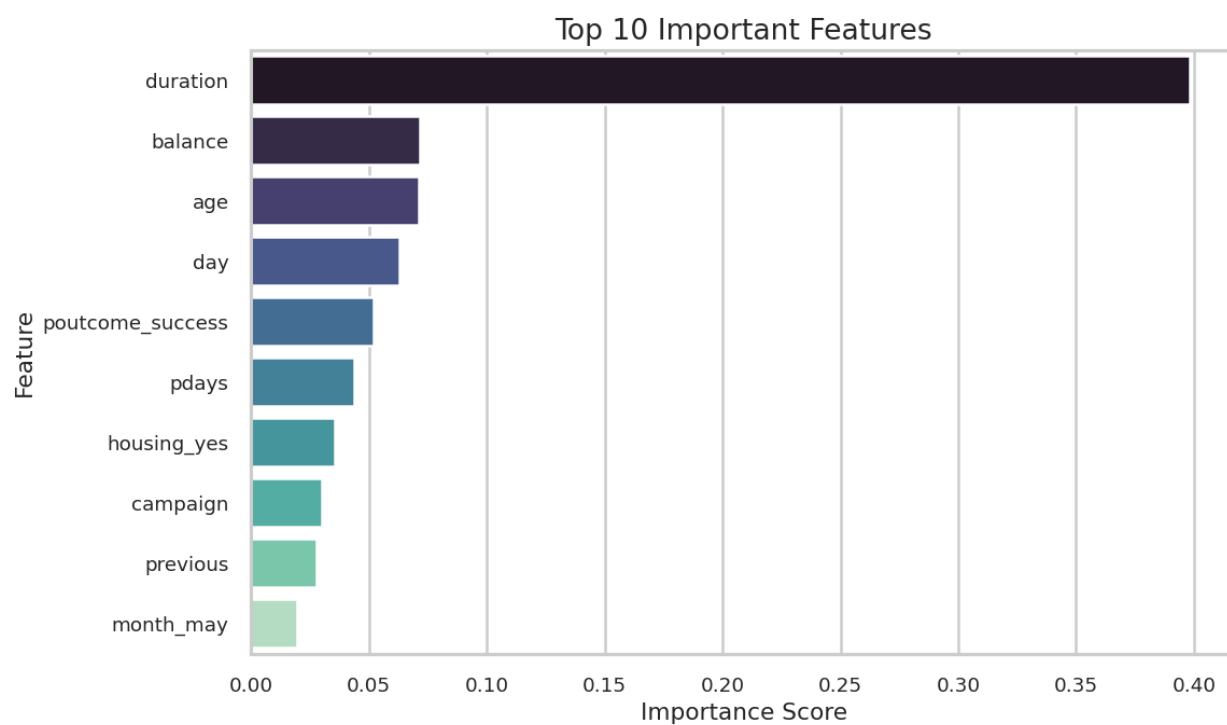


Figure 22: Top 10 Important Features

4.7 Precision-Recall Analysis

The Precision-Recall curve showed that the model easily beat the basic minimum score of 0.117, achieving an area under the curve of about 0.593. This proves that the model is genuinely good at spotting positive sign-ups, even when hidden inside highly uneven data.

Overall, the standard Random Forest model gave the most stable general results. However, reducing the size of the majority group proved to be the most effective way to boost recall and the F1-score, even if it caused a slight drop in total accuracy.

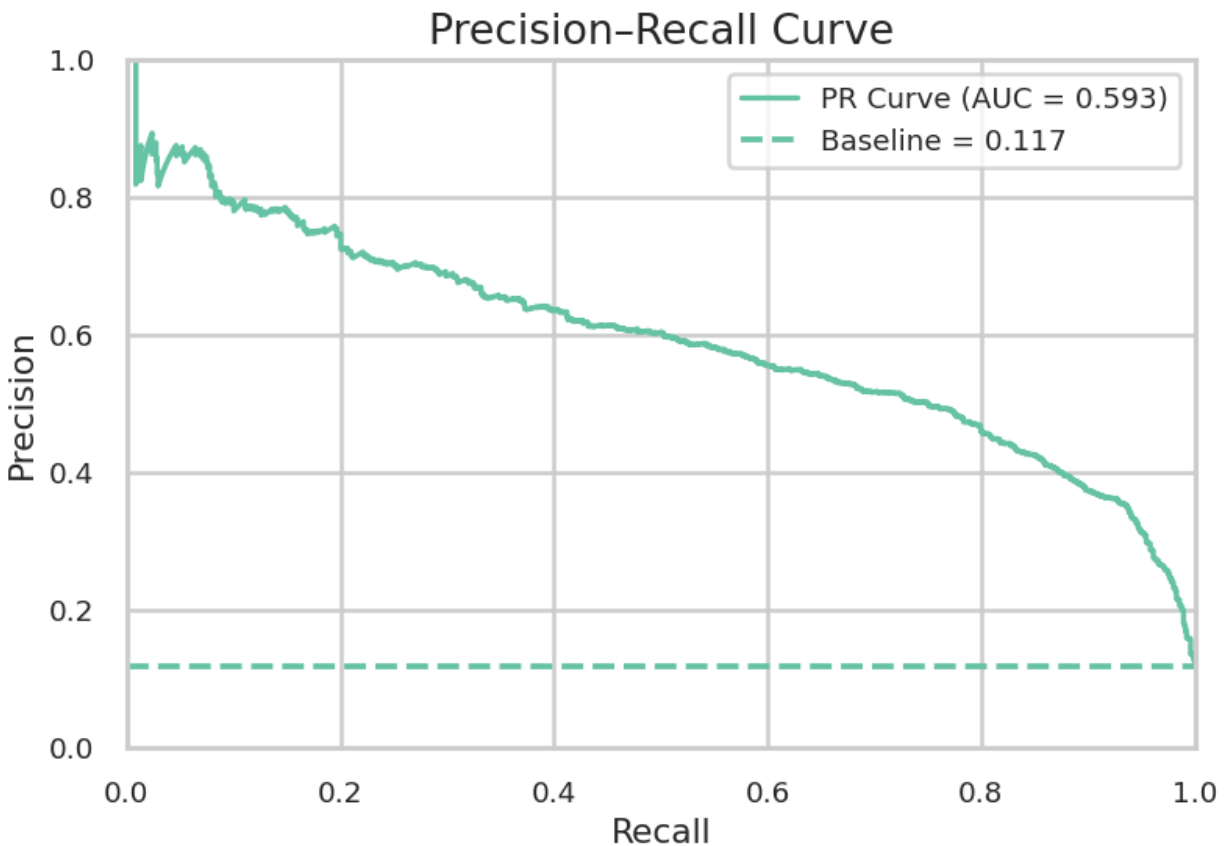


Figure 23: Precision - Recall Curve

5 Conclusion

This research aimed to predict whether a customer would agree to a term deposit by applying supervised machine learning to the Bank Marketing dataset. The project followed a complete analytical path, including cleaning the data, finding initial trends, adjusting the variables, and building the models. The dataset contained 45,211 records and 17 different details, but it suffered from a massive class imbalance, as only a very small number of clients actually accepted the offer.

Among the starting models, Random Forest produced the strongest all-around results, maintaining a stable balance between precision, recall, and ROC-AUC. While Logistic Regression achieved high overall accuracy, it failed miserably at finding the rare positive sign-ups. The Decision Tree model improved recall compared to Logistic Regression but was too unstable to rely on.

When addressing the uneven data, shrinking the majority group (undersampling) and using Random Forest proved highly effective for finding positive cases. This method boosted recall to 0.86 and earned the highest F1-score of 0.55, although it caused a drop in total accuracy and precision. Conversely, simply adjusting the class weights in the Random Forest slightly improved the ROC-AUC but failed to actually help the computer find more positive sign-ups.

These results prove that the best model depends entirely on what the business wants to achieve. If a bank wants stable, balanced predictions, the standard Random Forest is the best choice. However, if the main goal is to find every single potential customer even if it means making a few mistakes the undersampled model is much better because of its high recall.

5.1 Limitations

Several weaknesses in this study must be noted. First, the data contained hidden blank spaces labeled as "unknown," and the simple method used to fill them might have missed the true patterns in the data. Second, the extreme class imbalance makes it very difficult for the computer to accurately learn how to spot the rare "yes" answers.

Third, details like phone call duration create a problem called data leakage. This means the computer used a clue it could not possibly know in the real world (how long the call lasted) before the call actually happened, which likely made the model's scores look artificially high. Finally, the study only tested a small number of basic algorithms, ignoring more advanced options.

5.2 Future Work

Future research can fix these issues in several ways. Advanced tools for balancing data, such as SMOTE, could be used to create artificial "yes" answers and train the model better. More powerful algorithms, like XGBoost, LightGBM, or even deep learning, could be tested to see if they beat the current results.

Additionally, the model could be simplified by removing unhelpful details. It is also absolutely crucial to delete "cheating" variables, like call duration, so the model can function properly in the

real world. Finally, turning the model into a live software program and testing it on brand-new customer data would truly prove its value.

In summary, this research confirms that machine learning especially tools that combine many models together can successfully forecast customer choices and help banks run highly focused and profitable marketing campaigns.

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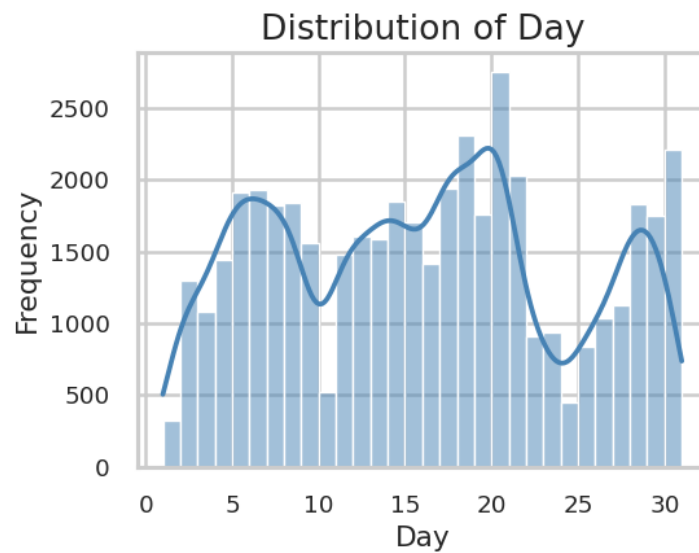
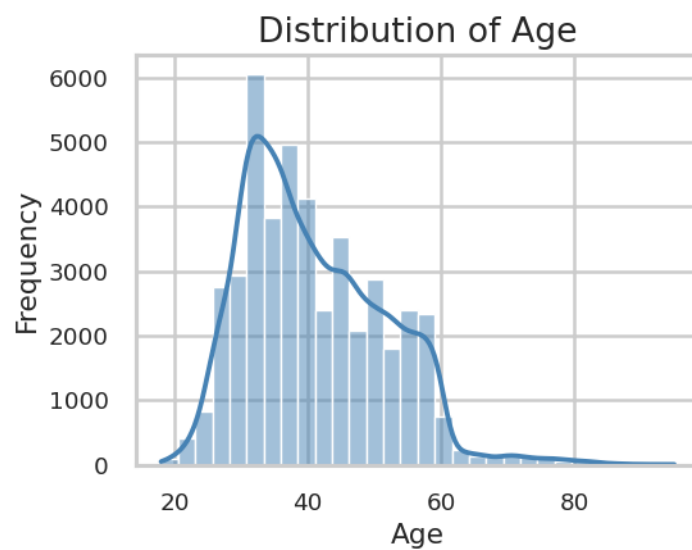
7 Appendix

Appendix A:

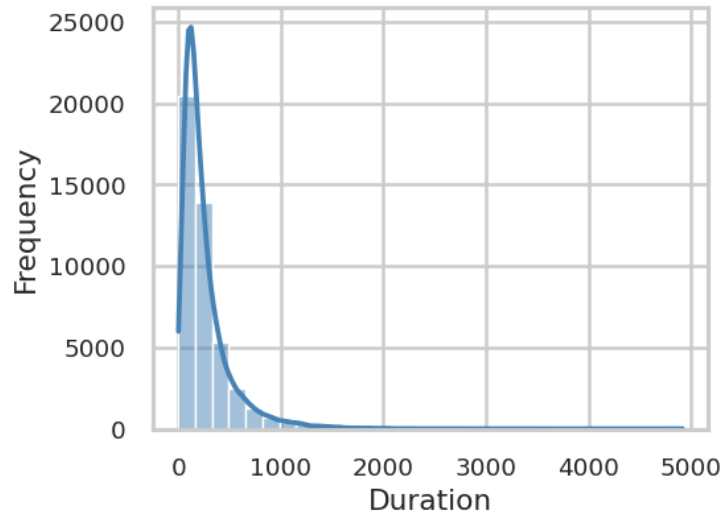
[Project Code](#)

Appendix B:

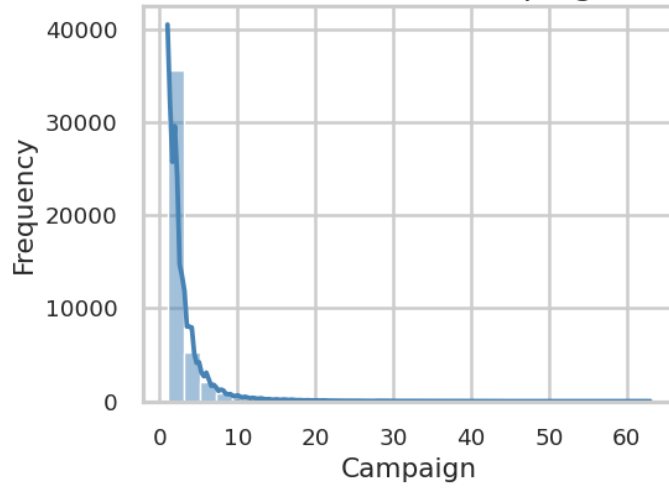
Variable Distributions.



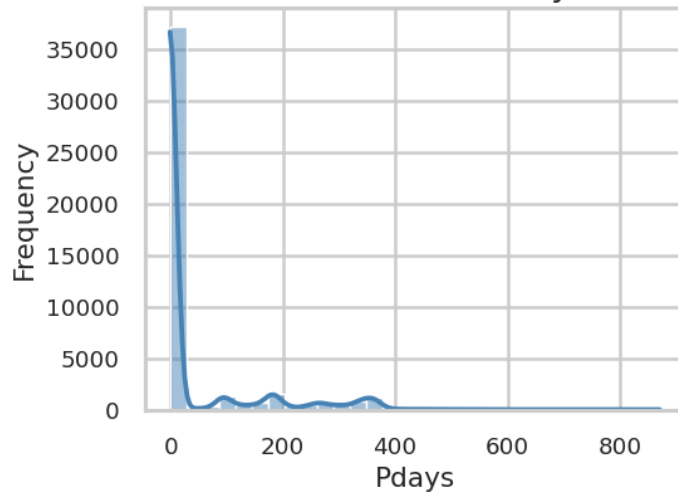
Distribution of Duration

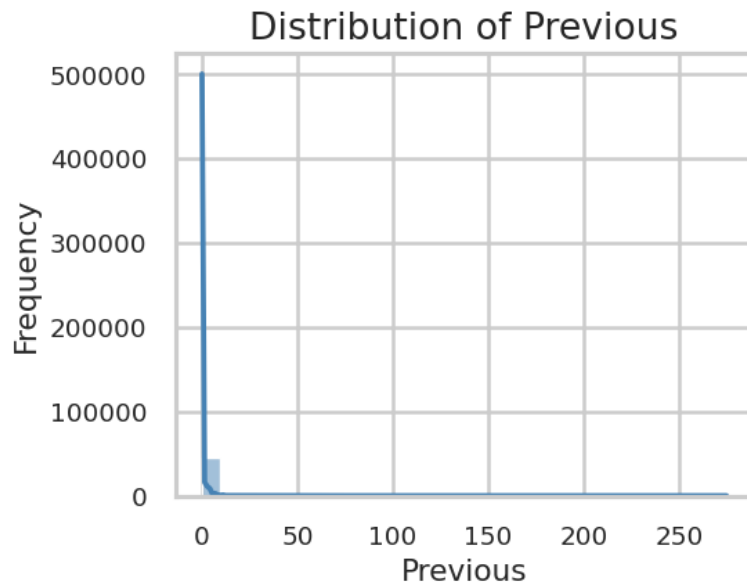


Distribution of Campaign



Distribution of Pdays

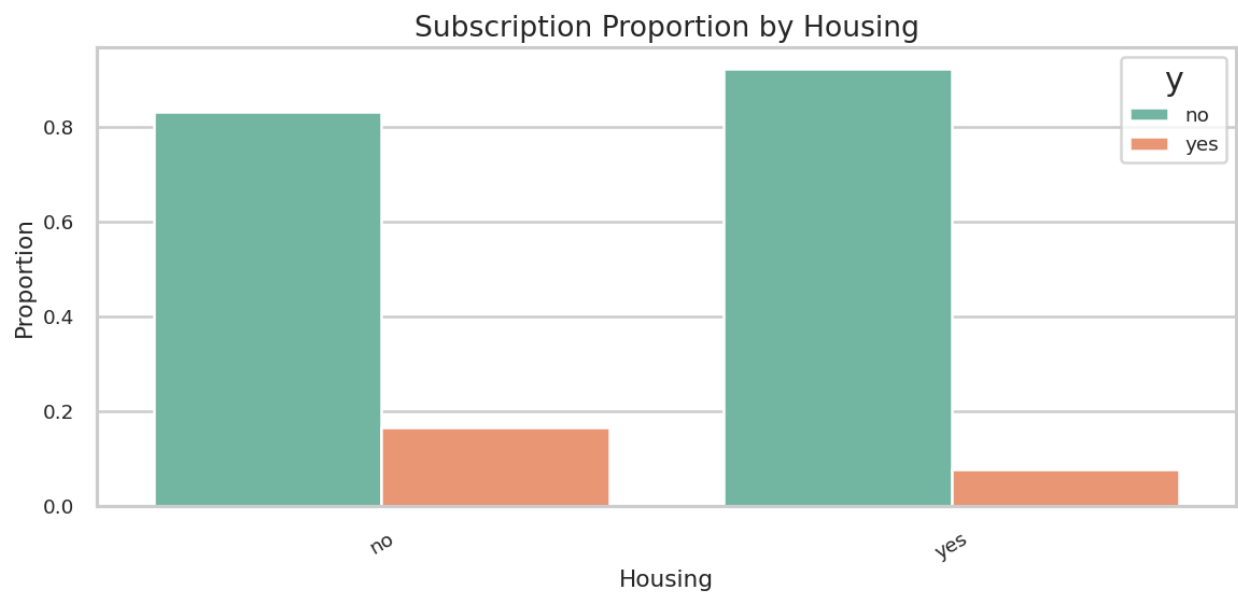


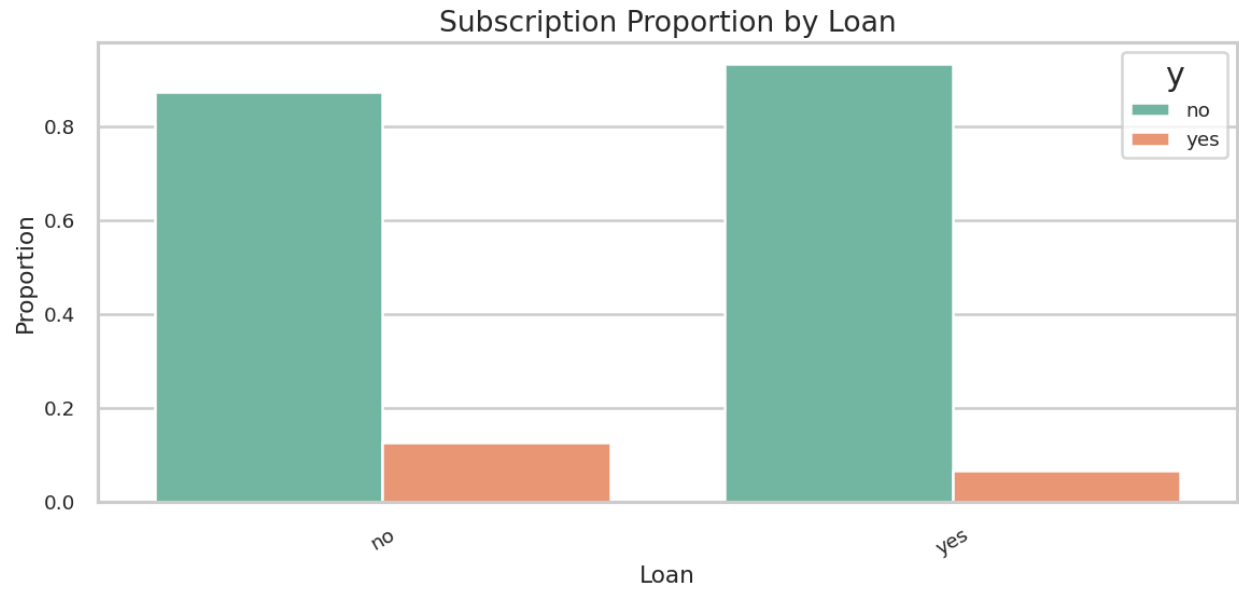


Appendix B images are the distribution plots of different independent variables in the dataset.

Appendix c:

Variable Subscription Proportions





Appendix C images are the bar plots of dependent variable with respect to Housing, Loan and Contact.